BIOS 6624: Project 3

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Load Data

```
fmhst_raw <- read_csv("/Users/ryan_summers/GitHub/BIOS6624-Class/Project3/DataRaw/frmgham2.csv")
```

```
## Rows: 11627 Columns: 39
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (39): RANDID, SEX, TOTCHOL, AGE, SYSBP, DIABP, CURSMOKE, CIGPDAY, BMI, D...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Data Management & Cleanup

Remove subjects who had event (stroke) at baseline and Filter to 10yrs cutoff. Calculate censored indicator if event doesn't happen within 10yrs and calculate time to censoring variable.

```
cutoff_days <- 10*365.25

fmhst_filt <- fmhst_raw %>%
  # remove subjects who had event prior to baseline assessment
  filter(TIMESTRK != 0 & PERIOD == 1) %>%   # baseline cohort
  mutate(
    # whether the subject was censored at the 10yr cutoff
    censored_at_10y = if_else(STROKE == 0 & TIMESTRK > cutoff_days, 1, 0),

    # follow-up time truncated at 10 years
    followup_10y = pmin(TIMESTRK, cutoff_days),

    # event indicator within 10 years
    STROKE_10y = if_else(STROKE == 1 & TIMESTRK <= cutoff_days, 1, 0),

    # time to censoring at the cutoff
    # only meaningful for those censored at 10 years
    time_to_censoring = if_else(censored_at_10y == 1, cutoff_days, NA_real_),

    # convert variables to factors
    SEX = factor(SEX, levels = c(1,2), labels = c("Men", "Women")),
    DIABETES = factor(DIABETES, levels = c(0,1), labels = c("No", "Yes")),
```

```
    CVD = factor(CVD, levels = c(0,1), labels = c("No", "Yes")),
    BPMEDS = factor(BPMEDS, levels = c(0,1), labels = c("No", "Yes")),
    CURSMOKE = factor(CURSMOKE, levels = c(0,1), labels = c("No", "Yes")),
    STROKE_10y = factor(STROKE_10y, levels = c(0,1), labels = c("No", "Yes")),
    censored_at_10y = factor(censored_at_10y, levels = c(0, 1), labels = c("No", "Yes"))
    )
```

We restricted the primary analysis to baseline risk factors measured at period 1 and modeled incident stroke over the first 10 years of follow-up. This approach aligns predictor measurement more closely with the risk window of interest. We additionally examined how major risk factors changed across the three examination periods to assess whether a time-varying covariate analysis might be informative. If risk factors change materially over time, a baseline-only model may understate or misclassify the relationship between current risk-factor status and stroke hazard. However, time-varying covariate analyses in observational data may also be harder to interpret because updated covariates can be influenced by treatment changes and evolving disease status.

Exploratory Data Analysis (EDA)

Table1

```
tbl1 <- fmhst_filt %>%
  tbl_summary(
    by = SEX,
    include = c(
      AGE, SYSBP, DIABETES, TOTCHOL, CVD, BPMEDS, CURSMOKE,
      BMI, STROKE_10y, censored_at_10y),
    type = list(
      all_continuous() ~ "continuous2",
      all_categorical() ~ "categorical"),
    statistic = list(
      all_continuous() ~ c(
        "{mean} ({sd})",
        "{median} ({p25}, {p75})",
        "{min} - {max}"),
      all_categorical() ~ "{n} ({p}%)"),
    digits = all_continuous() ~ 1,
    missing = "ifany",
    missing_text = "NA",
    missing_stat = "{N_miss} ({p_miss}%)",
    label = list(
      AGE ~ "Age",
      SYSBP ~ "Systolic BP",
      DIABETES ~ "Diabetes",
      TOTCHOL ~ "Serum Total Cholesterol (mg/dL)",
      CVD ~ "Cardiovascular Disease",
      BPMEDS ~ "BP Meds",
      CURSMOKE ~ "Smoker",
      STROKE_10y ~ "Event (Stroke)",
      censored_at_10y ~ "Censored")) %>%
  add_overall() %>% bold_labels() %>% italicize_levels() %>% add_n() %>%
  as_gt() %>%
```

```r
  gt::fmt_markdown(columns = everything()) %>%
  tab_options(
    table.font.size = "small",
    heading.align = "left",
    data_row.padding = px(3),
    table.border.top.width = px(2),
    table.border.bottom.width = px(2))

tbl1
```

```r
# number of participants
cat("Number of participants in entire cohort",
   length(unique(fmhst_raw$RANDID)), "\n\n")
```

```
## Number of participants in entire cohort 4434
```

```r
cat("Number of participants in 10yr subset cohort",
   length(unique(fmhst_filt$RANDID)), "\n\n")
```

```
## Number of participants in 10yr subset cohort 4402
```

Characteristic	N	Overall N = 4,402[1]	Men N = 1,930[1]	Women N = 2,472[1]
Age	4,402			
Mean (SD)		49.9 (8.7)	49.7 (8.7)	50.0 (8.6)
Median (Q1, Q3)		49.0 (42.0, 57.0)	49.0 (42.0, 57.0)	49.0 (43.0, 57.0)
Min - Max		32.0 - 70.0	33.0 - 69.0	32.0 - 70.0
Systolic BP	4,402			
Mean (SD)		132.8 (22.3)	131.6 (19.4)	133.7 (24.3)
Median (Q1, Q3)		128.5 (117.0, 144.0)	128.5 (118.0, 141.0)	128.5 (116.0, 146.0)
Min - Max		83.5 - 295.0	83.5 - 235.0	83.5 - 295.0
Diabetes	4,402			
No		4,283 (97%)	1,871 (97%)	2,412 (98%)
Yes		119 (2.7%)	59 (3.1%)	60 (2.4%)
Serum Total Cholesterol (mg/dL)	4,350			
Mean (SD)		237.0 (44.6)	233.6 (42.4)	239.7 (46.2)
Median (Q1, Q3)		234.0 (206.0, 264.0)	231.0 (206.0, 259.0)	237.0 (206.0, 268.0)
Min - Max		113.0 - 696.0	113.0 - 696.0	129.0 - 600.0
NA		52 (1.2%)	7 (0.4%)	45 (1.8%)
Cardiovascular Disease	4,402			
No		3,277 (74%)	1,258 (65%)	2,019 (82%)
Yes		1,125 (26%)	672 (35%)	453 (18%)
BP Meds	4,342			
No		4,206 (97%)	1,868 (98%)	2,338 (96%)
Yes		136 (3.1%)	40 (2.1%)	96 (3.9%)
NA		60 (1.4%)	22 (1.1%)	38 (1.5%)
Smoker	4,402			
No		2,233 (51%)	762 (39%)	1,471 (60%)
Yes		2,169 (49%)	1,168 (61%)	1,001 (40%)
BMI	4,385			
Mean (SD)		25.8 (4.1)	26.2 (3.4)	25.6 (4.5)
Median (Q1, Q3)		25.5 (23.1, 28.1)	26.1 (24.0, 28.3)	24.8 (22.5, 27.8)
Min - Max		15.5 - 56.8	15.5 - 40.4	16.0 - 56.8
NA		17 (0.4%)	4 (0.2%)	13 (0.5%)
Event (Stroke)	4,402			
No		4,291 (97%)	1,881 (97%)	2,410 (97%)
Yes		111 (2.5%)	49 (2.5%)	62 (2.5%)
Censored	4,402			
No		762 (17%)	391 (20%)	371 (15%)
Yes		3,640 (83%)	1,539 (80%)	2,101 (85%)

[1] n (%)